Joint symmetry and dynamical accessibility in compact Hamiltonian encodings of set cover

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Which part of a Hamiltonian spectrum is physically relevant when both the initial state and interpolation preserve several symmetries? We address this question for a compact multi-register encoding of Minimum Set Cover. The represented symmetry combines register permutations with the faithful base action of the incidence automorphism group. We distinguish its fixed, symmetry-allowed space from the generally smaller cyclic space generated by the protocol and define gaps relative to isolated bands in that space. A sector-resolved Schur-complement bound certifies cover-measurement probability whenever the sector contains feasible covers and has positive invalid-state separation; the sector kinetic floor is essential. Exact orbit quotients expose inter-sector coincidences invisible to a symmetric protocol, while a stability theorem shows that transverse dark crossings persist under small symmetry-preserving perturbations. On a family of even cycles with n vertices, the original linear interpolation has an exact global multiplicity closure while the joint-fixed excitation gap remains constant. For the same family we construct a Johnson/Metropolis parent path from a Dicke state to a Gibbs-amplitude state whose cover-failure probability is bounded by a constant times n^{-5}. A zero-range comparison and a log-concave three-box coupling give the explicit uniform cyclic-gap lower bound $1024\,n^{-13}$ along the path. Quantitative derivative bounds then yield a polynomial adiabatic runtime in the abstract Hamiltonian-access model, conditional on Dicke-state preparation and access to the parent Hamiltonian. We make no quantum-speedup claim: the result is a rigorous separation of global, symmetry-allowed, and dynamically accessible spectral structure. Changing the initial state or breaking a preserved symmetry changes that accessible spectrum.

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1 Introduction

Symmetry block diagonalizes a Hamiltonian, but block diagonalization alone does not determine which eigenstates a protocol can occupy. If an initial state is fixed by every preserved symmetry, evolution is confined to the joint fixed space. Even that statement can be loose: the algebra generated by the endpoint Hamiltonians may act cyclically only on a proper subspace of the joint fixed space. A small global gap can then describe a crossing that the protocol neither couples to nor detects.

Set Cover admits the usual one-binary-variable-per-base Ising formulation, and it also appears among the constrained examples motivating quantum alternating-operator ansatzes [1, 2]. Separately, qubit-efficient variational encodings show that logarithmic or otherwise compressed register counts are not by themselves a novelty claim [3–5]. Our multi-register address encoding belongs to this broad compact-encoding line; the contribution claimed here is spectral rather than a new qubit count.

Classical objective symmetries can confine quantum approximate optimization algorithm (QAOA) dynamics and impose equal measurement probabilities on symmetry-related strings [6]. Quantum walks on automorphism quotients likewise identify a smaller invariant evolution for suitable initial states [7], while Krylov methods use repeated generator action to identify a minimal dynamical subspace [8]. These ideas motivate, but do not identify with one another, the two spaces used below: the group-fixed orbit space and the generally smaller cyclic space of the full protocol algebra.

Finally, symmetric discriminants of reversible Markov chains and annealing paths with Gibbs-amplitude ground states are established tools in quantum walk and quantum simulated-annealing constructions [9, 10]. Quantitative adiabatic statements require an isolated spectral band together with derivative and gap control [11]. We use the discriminant only to construct a rigorously gapped witness path and make no speedup claim.

Minimum Set Cover (MSC) provides a concrete setting in which these three spaces differ. With n_B candidate bases, a k-register encoding represents an ordered candidate by $k\lceil\log_2 n_B\rceil$ address qubits. Permuting the registers gives an action of the symmetric group S_k, while colour-preserving automorphisms of the incidence relation permute the base labels. Register symmetry alone and a final-Hamiltonian sector gap are insufficient for an adiabatic or otherwise path-dependent statement: one must use the faithful joint action, the protocol's cyclic space, and the minimum isolation gap along an explicit path.

This work makes that distinction its central question. In the summary below, G_B denotes the faithfully represented group of base automorphisms, $\mathcal{H}_{\mathrm{sym}}$ the subspace fixed by $S_k \times G_B$, and $\mathcal{K}_u$ the cyclic space generated from the uniform initial state by the endpoint Hamiltonians. Their precise definitions are given in Section 2. The three main contributions are:

1. the faithful joint symmetry $S_k \times G_B$, exact orbit quotients, and a strict distinction between the symmetry-allowed space $\mathcal{H}_{\mathrm{sym}}$ and the protocol-cyclic space $\mathcal{K}_u$;
2. a sector-localization theorem with the necessary kinetic floor, together with a gap taxonomy and stable inter-sector crossings that are globally visible but dynamically dark; and
3. an even-cycle family with an exact dark multiplicity closure and a distinct parent path whose accessible gap is uniformly polynomial, yielding a conditional polynomial-time adiabatic corollary.

The last result is purposely about accessibility rather than computational advantage. The controlled cycle family is classically easy, and the proven exponent is conservative. Its role is to separate the gap question cleanly into symmetry, geometry, and kinetics, without inferring asymptotic behavior from finite-size data.

Neither orbit reduction, compact addressing, nor a Markov discriminant is claimed as new in isolation. The contribution is their use in a single protocol-dependent theory that separates global, group-fixed, and cyclic spectra and proves that these distinctions persist in an analytically controlled family.

We use the following notation throughout. For an operator X, $\operatorname{Ran} X$ and $\operatorname{spec} X$ denote its range and spectrum. The symbol $\|\cdot\|$ denotes the Hilbert-space norm on vectors and the induced operator norm on operators. The gap of a finite-dimensional Hermitian operator is the distance from its lowest eigenspace to the next distinct eigenvalue; $\lambda_{\min}(X|_{\mathcal{L}})$ is the lowest eigenvalue of the compression of X to $\mathcal{L}$. For positive g, the notation $f = O(g)$ means $|f| \le Cg$ for a constant $C > 0$ independent of the asymptotic variable in the stated limit.

2 Encoding, joint symmetry, and cyclic accessibility

Let n_B and n_T be the numbers of bases and tasks, respectively, and fix a positive integer k of address registers. The incidence matrix is $A \in \{0,1\}^{n_B \times n_T}$, with $A_{bt} = 1$ when base b covers task t. We first work in the physical address space

$$\mathcal{H}_{\mathrm{valid}} = (\mathbb{C}^{n_B})^{\otimes k}; \tag{1}$$

the embedding into $2^{\lceil \log_2 n_B \rceil}$ levels and its dynamically disconnected padding block are implementation details.

Write $|\boldsymbol{b}\rangle = |b_1\rangle \otimes \cdots \otimes |b_k\rangle$ for a computational-basis tuple. Its numbers of uncovered tasks and repeated base pairs are

$$U(\boldsymbol{b}) = \sum_{t=1}^{n_T} \prod_{r=1}^{k} (1 - A_{b_r,t}), \tag{2}$$

$$D(\boldsymbol{b}) = \sum_{1 \le r < s \le k} \mathbf{1}[b_r = b_s], \tag{3}$$

respectively, where $\mathbf{1}[\cdot]$ is the indicator of the enclosed condition. Let $|e\rangle = \sum_{b=1}^{n_B} |b\rangle$ and let I_B be the identity on one address register. The normalized complete-graph hopping matrix and its k-register lift are

$$W = \frac{|e\rangle\langle e| - I_B}{n_B - 1}, \qquad H_{\mathrm{walk}} = \sum_{r=1}^{k} W^{(r)}, \tag{4}$$

where $W^{(r)}$ acts as W on register r and as the identity on all others. Define the diagonal penalty operators

$$H_{\mathrm{cov}} = \sum_{\boldsymbol{b}} U(\boldsymbol{b}) |\boldsymbol{b}\rangle\langle \boldsymbol{b}|, \qquad H_{\mathrm{excl}} = \sum_{\boldsymbol{b}} D(\boldsymbol{b}) |\boldsymbol{b}\rangle\langle \boldsymbol{b}|. \tag{5}$$

Both sums run over all n_B^k computational-basis tuples. For positive penalty energies λ and μ, the problem Hamiltonian is

$$H_{\mathrm{prob}} = H_{\mathrm{walk}} + \lambda H_{\mathrm{cov}} + \mu H_{\mathrm{excl}}, \tag{6}$$

so $\lambda U(\boldsymbol{b}) + \mu D(\boldsymbol{b})$ is the total diagonal penalty of $|\boldsymbol{b}\rangle$. An explicit initial Hamiltonian is

$$H_{\mathrm{init}} = \sum_{r=1}^{k} \big(I_B - |u_B\rangle\langle u_B|\big)_r, \qquad |u_B\rangle = n_B^{-1/2} \sum_{b=1}^{n_B} |b\rangle, \tag{7}$$

with unique ground state $|u\rangle = |u_B\rangle^{\otimes k}$ and initial gap one. The original linear interpolation is parameterized by $s \in [0,1]$ as

$$H(s) = (1-s) H_{\mathrm{init}} + s H_{\mathrm{prob}}. \tag{8}$$

2.1 The faithfully represented group

The colour-preserving incidence automorphism group is

$$G_A = \{(\pi_B, \pi_T) \in S_{n_B} \times S_{n_T} : A_{\pi_B(b),\pi_T(t)} = A_{bt} \text{ for all } b, t\}, \tag{9}$$

where S_d denotes the symmetric group on d labels. Only its base component acts on Eq. (1). We therefore use the effective group

$$G_B = \mathrm{pr}_B(G_A) \simeq G_A/\ker\rho, \qquad \rho(\pi_B, \pi_T) = \pi_B, \tag{10}$$

where pr_B projects an automorphism onto its base permutation and $\ker\rho$ contains the automorphisms acting trivially on all bases. Thus G_B is the faithfully represented base group. The register and base actions commute, so the represented symmetry is

$$G = S_k \times G_B. \tag{11}$$

Every term in Eqs. (6) and (7) commutes with G. Consequently, any Hamiltonian H commuting with G has the block structure

$$\mathcal{H}_{\text{valid}} \simeq \bigoplus_{\alpha,\beta} V_\alpha^{S_k} \otimes V_\beta^{G_B} \otimes \mathcal{M}_{\alpha\beta}, \tag{12}$$

$$H \simeq \bigoplus_{\alpha,\beta} I_{V_\alpha} \otimes I_{V_\beta} \otimes h_{\alpha\beta}. \tag{13}$$

Here α and β label irreducible representations of S_k and G_B; the spaces $V_\alpha^{S_k}$ and $V_\beta^{G_B}$ carry those representations; $\mathcal{M}_{\alpha\beta}$ is their multiplicity space; and $h_{\alpha\beta}$ is the reduced action on that space. The symbols I_{V_α} and I_{V_β} denote the corresponding identities. This is the source of Schur multiplicities; an equality of eigenvalues in different blocks is a separate phenomenon. The full representation and the role of the kernel of G_A are detailed in Appendix B.

2.2 Allowed space versus accessible space

Let U_σ permute the k registers according to $\sigma \in S_k$, and let V_g apply the base permutation $g \in G_B$ in every register. The joint fixed projector and its range are

$$\Pi_{\text{sym}} = \left(\frac{1}{k!}\sum_{\sigma\in S_k} U_\sigma\right)\left(\frac{1}{|G_B|}\sum_{g\in G_B} V_g\right), \tag{14}$$

$$\mathcal{H}_{\text{sym}} = \operatorname{Ran}\Pi_{\text{sym}}. \tag{15}$$

This is the symmetry-allowed invariant space, not automatically the space filled by the protocol. Let $\mathfrak{A} = \mathrm{alg}^*(H_{\text{init}}, H_{\text{prob}}, I)$ be the smallest unital algebra containing the endpoints and closed under adjoints; here I is the identity on $\mathcal{H}_{\text{valid}}$. The dynamically accessible cyclic space is

$$\begin{aligned}\mathcal{K}_u &= \mathfrak{A}|u\rangle \\ &= \mathrm{span}\{X_\ell \cdots X_1|u\rangle : X_i \in \{H_{\text{init}}, H_{\text{prob}}\}, \ell \geq 0\}.\end{aligned} \tag{16}$$

It is the smallest common reducing subspace that contains the initial state, and in general

$$\mathcal{K}_u \subseteq \mathcal{H}_{\text{sym}} \subseteq \mathcal{H}_{\text{valid}}. \tag{17}$$

The first inclusion can be strict; padding makes it strict automatically if one defines the symmetry on the full qubit space.

Figure 1 summarizes the two distinctions used throughout the paper. Group invariance identifies an allowed space, while the initial state and generator algebra select the smaller cyclic space. An eigenvalue coincidence in another irrep may then close the global multiplicity gap without closing the isolation gap seen by the protocol.

2.3 Exact orbit quotient

Orbit quotients of symmetry-confined quantum walks are established constructions [7]; we use the joint action here to retain the exact Hamiltonian matrix elements relevant to the present encoding. For a G-orbit $\mathcal{O}$ in the computational basis, define $|\mathcal{O}\rangle = |\mathcal{O}|^{-1/2}\sum_{x\in\mathcal{O}}|x\rangle$. These orbit states are an orthonormal basis of $\mathcal{H}_{\text{sym}}$. If H is invariant, $x \in \mathcal{O}$, and $H_{yx} := \langle y|H|x\rangle$, then

$$\langle\mathcal{O}'|H|\mathcal{O}\rangle = \sqrt{\frac{|\mathcal{O}|}{|\mathcal{O}'|}}\sum_{y\in\mathcal{O}'} H_{yx}. \tag{18}$$

The sum is independent of the representative. The orbit-counting (Burnside) lemma [12] gives

$$\dim\mathcal{H}_{\text{sym}} = \frac{1}{|G_B|}\sum_{g\in G_B}[z^k]\prod_{\ell\geq 1}(1-z^\ell)^{-c_\ell(g)}, \tag{19}$$

where G_B is the faithful base group in Eq. (10), $[z^k]$ extracts the coefficient of z^k, and $c_\ell(g)$ counts length-ℓ cycles in the base action. Orbit states need not span $\mathcal{K}_u$ minimally; cyclic closure must be tested separately. Appendix B derives Eq. (19) from the joint action.

3 Sector localization and relevant gaps

Let P project onto the span of computational-basis tuples that use distinct bases and cover every task, and put $Q = I - P$. Write $V = \lambda H_{\text{cov}} + \mu H_{\text{excl}}$, $m = \min\{\lambda, \mu\}$ for the minimum single-constraint penalty, and $\kappa = k/(n_B - 1)$ for the magnitude of the lowest possible k-register walk energy. Here k is

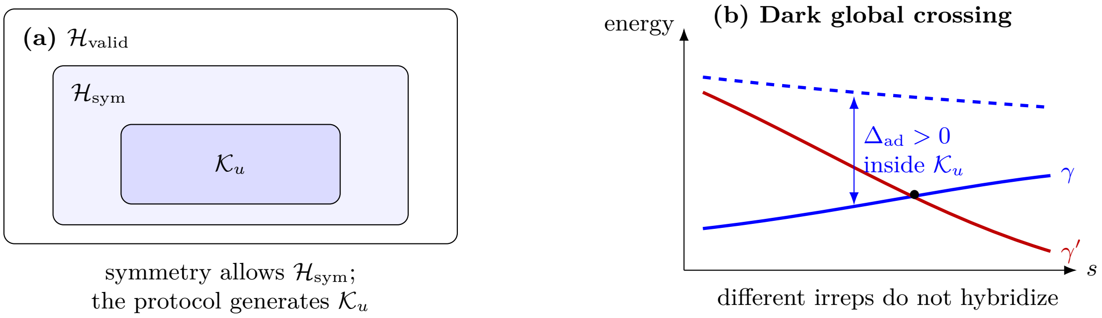

Figure 1: Protocol-dependent spectral relevance. (a) The joint-fixed space is only symmetry allowed; the cyclic space also depends on the initial state and generated algebra. (b) The labels γ and γ' denote inequivalent symmetry sectors. A branch in the latter can cross the followed branch and close a global multiplicity gap while the tracked band remains isolated within the cyclic space.

the register count fixed in Section 2. Then $VP = 0$ and the complete-graph normalization gives

$$QH_{\text{prob}}Q \geq (m - \kappa)Q. \tag{20}$$

The global localization statement cannot be copied verbatim into an arbitrary sector. A projected cover is a superposition of computational covers and can have nonzero kinetic expectation. The missing term is the sector's kinetic floor.

Proposition 1 (Sector localization)**.** *Let γ label a symmetry sector and let Π_γ be its projector, commuting with H_{prob}, H_{walk}, V, and P, and suppose $P_\gamma = P\Pi_\gamma \neq 0$. Define the invalid-sector projector Q_γ, the feasible-to-invalid coupling B_γ, and the sector kinetic floor η_γ by*

$$Q_\gamma = Q\Pi_\gamma, \qquad B_\gamma = Q_\gamma H_{\text{walk}} P_\gamma, \tag{21}$$
$$\eta_\gamma = \lambda_{\min}\left(P_\gamma H_{\text{walk}} P_\gamma|_{\operatorname{Ran} P_\gamma}\right). \tag{22}$$

If $Q_\gamma = 0$, every state in the sector lies in $\operatorname{Ran} P$ *and the leakage is identically zero. Otherwise, let E_0^γ be the lowest energy in the sector and define the compressed invalid-state separation*

$$\Gamma_\gamma = \lambda_{\min}\left(Q_\gamma H_{\text{prob}} Q_\gamma|_{\operatorname{Ran} Q_\gamma}\right) - E_0^\gamma. \tag{23}$$

If $\Gamma_\gamma > 0$, every normalized sector ground state ψ_γ satisfies

$$1 - \langle\psi_\gamma|P|\psi_\gamma\rangle \leq \frac{\|B_\gamma\|^2}{\Gamma_\gamma^2 + \|B_\gamma\|^2}. \tag{24}$$

Moreover,

$$\Gamma_\gamma \geq m - \kappa - \eta_\gamma, \tag{25}$$

so the right-hand side of Eq. (24) remains valid with Γ_γ replaced by $g_\gamma = m - \kappa - \eta_\gamma > 0$.

Physically, the certificate compares the energetic separation Γ_γ protecting the feasible sector with the coupling $\|B_\gamma\|$ that leaks amplitude into invalid states. The kinetic floor η_γ accounts for delocalization within the chosen symmetry sector. The full proof and the cyclic-space caveat are in Appendix A.

This statement certifies the solution content of the lowest-energy state within a sector. It does not say that the ground vector has exact support on $\operatorname{Ran} P$: for finite penalties Eq. (24) controls, rather than eliminates, leakage. Exact support occurs only in an infinite penalty limit or in special decoupled cases.

The theorem applies to genuine commuting symmetry sectors. A cyclic projector need not commute with P, so cyclic-space solution probability is evaluated directly or through a separately constructed commuting projector.

3.1 Protocol-dependent gaps and conditional no-go statements

For an invariant space $\mathcal{L}$, let $P_0^{\mathcal{L}}(s)$ be the ground projector of $H(s)|_{\mathcal{L}}$ and $g_0^{\mathcal{L}}(s) = \operatorname{rank} P_0^{\mathcal{L}}(s)$. Write the eigenvalues of the restriction, ordered with multiplicity, as $E_0^{\mathcal{L}}(s) \leq E_1^{\mathcal{L}}(s) \leq \cdots$. We keep separate

$$\delta_{\text{mult}}^{\mathcal{L}}(s) = E_1^{\mathcal{L}}(s) - E_0^{\mathcal{L}}(s), \tag{26}$$
$$\Delta_{\text{exc}}^{\mathcal{L}}(s) = E_{g_0^{\mathcal{L}}(s)}^{\mathcal{L}}(s) - E_0^{\mathcal{L}}(s). \tag{27}$$

The first vanishes whenever the ground state is degenerate; the second is the excitation gap above the whole manifold. A Schur multiplicity or an exact coincidence between irreps is therefore not called a gap closure when Eq. (27)'s second quantity stays positive.

Let E_0, E_0^{sym}, and $E_0^{\mathcal{K}}$ be the ground energies of $H(s)$ on $\mathcal{H}_{\text{valid}}$, $\mathcal{H}_{\text{sym}}$, and $\mathcal{K}_u$, respectively, at a common value of s. The nested spaces in Eq. (17) define the accessibility costs

$$\delta_{\text{sym}} = E_0^{\text{sym}} - E_0, \qquad \delta_{\text{cyc}} = E_0^{\mathcal{K}} - E_0^{\text{sym}}, \tag{28}$$
$$\delta_{\text{acc}} = E_0^{\mathcal{K}} - E_0 = \delta_{\text{sym}} + \delta_{\text{cyc}}. \tag{29}$$

A positive δ_{acc} says that the protocol follows a globally excited branch. That branch can nevertheless have high cover probability.

For adiabatic evolution, the relevant object is an isolated spectral band, not merely the first distinct final eigenvalue [11]. Let $P_{\mathrm{track}}(s)$ be a continuous isolated spectral projector on $\mathcal{K}_u$ whose range contains the initial ground state. Set $H_{\mathcal{K}_u}(s) = H(s)|_{\mathcal{K}_u}$ and let $H_{\mathrm{track}}(s)$ be the compression of $H_{\mathcal{K}_u}(s)$ to $\operatorname{Ran} P_{\mathrm{track}}(s)$. For finite real sets A, B, write $\operatorname{dist}(A, B) = \min_{a\in A, b\in B} |a - b|$. Define

$$\Delta_{\mathrm{ad}}(s) = \operatorname{dist}(\operatorname{spec} H_{\mathrm{track}}(s), \operatorname{spec} H_{\mathcal{K}_u}(s) \setminus \operatorname{spec} H_{\mathrm{track}}(s)). \quad (30)$$

An isolated continuous projector has constant rank. If its intended rank changes, an isolation gap has closed and a nondegenerate adiabatic formula cannot simply be continued through that point.

The four quantities have the following physical meanings:

quantity	physical meaning
δ_{mult}	splitting inside the lowest manifold; zero for a degenerate ground energy
Δ_{exc}	excitation gap above the entire lowest manifold
δ_{acc}	energy cost of confinement to the protocol-cyclic space
Δ_{ad}	isolation of the spectral band followed inside that space

What fixed-Hamiltonian evolution does not prove. For a time-independent Hamiltonian $H = \sum_j E_j P_j$, where P_j projects onto the eigenspace of the distinct energy E_j, unitary evolution from $\psi(0)$ preserves every energy-eigenspace population,

$$\|P_j\psi(t)\|^2 = \|P_j\psi(0)\|^2. \quad (31)$$

This rules out closed-system relaxation to a different eigenstate and fixes the diagonal ensemble. It does *not* imply that the probability of a computationally marked subspace is constant: interference between distinct energies can create transient concentration. A finite time scan cannot prove that no later time does better.

The statement is also initial-state dependent. A solution eigenstate used as the initial state already has unit solution population; a state with appropriate coherences can reach a marked subspace transiently. The exact symmetry obstruction is narrower and stronger:

Lemma 2 (Symmetry-conditioned inaccessibility)**.** *If $[H(t), \Pi_\gamma] = 0$ for every t and $\Pi_\gamma\psi(0) = \psi(0)$, then $\Pi_\gamma\psi(t) = \psi(t)$ for all t. Any target vector orthogonal to $\operatorname{Ran}\Pi_\gamma$ has exactly zero transition amplitude.*

Thus changing the initial state to one adapted to another irrep, or adding a generator with nonzero inter-sector matrix elements, changes the accessible spectrum and can evade the obstruction. This is why our no-go claims are always indexed by an initial state, a generator algebra, and the symmetries preserved by the full path.

4 Dark crossings in exact symmetry quotients

If Π_a and Π_b project onto inequivalent irreps of a symmetry preserved by the entire path, then

$$\Pi_a\, \partial_s H(s)\, \Pi_b = 0. \quad (32)$$

An equality of the two branch energies is therefore a global crossing but not an avoided crossing seen by a state confined to either block.

This phenomenon is stable within the symmetry-preserving class; it is not restricted to exactly solvable diagonal cancellation points.

Theorem 3 (Stability of a transverse dark crossing)**.** *Let $H_\varepsilon(s) = H(s) + \varepsilon R(s)$ be a continuously differentiable finite-dimensional path, where $\varepsilon \in \mathbb{R}$ is the perturbation strength and $R(s)$ is Hermitian. Let Π_a, Π_b project onto inequivalent symmetry sectors. Suppose every $H_\varepsilon(s)$ commutes with both projectors. Assume that near $(s_\star, 0)$ the two restricted blocks have simple isolated eigenvalue branches $E_a(s, \varepsilon)$ and $E_b(s, \varepsilon)$ such that*

$$E_a(s_\star, 0) = E_b(s_\star, 0), \qquad \partial_s(E_a - E_b)(s_\star, 0) \neq 0. \quad (33)$$

Then, for all sufficiently small $|\varepsilon|$, there is a unique nearby $s_\star(\varepsilon) = s_\star + O(|\varepsilon|)$ at which the two branches cross exactly, and

$$\Pi_a \partial_s H_\varepsilon(s) \Pi_b = 0. \quad (34)$$

If the a branch is separated from the rest of its block by at least $\Delta_\star > 0$ throughout a neighbourhood of $s_\star$ at $\varepsilon = 0$, then its isolation gap remains at least $\Delta_\star/2$ whenever $|\varepsilon| \sup_{s\in[0,1]} \|R(s)\| \le \Delta_\star/4$ and the perturbed crossing stays in that neighbourhood.

Physically, the preserved symmetry forbids the two branches from hybridizing. A symmetry-preserving perturbation may move the crossing, but cannot open an avoided crossing between the sectors. The complete stability argument is in Appendix C.1.

4.1 A finite joint-symmetry audit

For a frozen 2×4 incidence instance with $k = 3$, $\lambda = 5$, and $\mu = 10$, $G_B \simeq D_4$, where D_4 is the order-eight dihedral group, and the $S_3 \times D_4$ fixed quotient has dimension 22, compared with 512 in the valid ordered space. Cyclic closure from $|u\rangle$ also has dimension 22, as determined independently by the generator algebra. The uncatalysed path has a fixed-space rank closure at $s = 7/15$, showing that joint symmetry alone does not guarantee a rank-one accessible gap.

As a finite illustration of a dark crossing away from the cancellation point, use the single-register walk W from Eq. (4) and define

$$C_2 = -\sum_{1\le r<t\le k} W^{(r)}W^{(t)}, \tag{35}$$

and use the endpoint-preserving path

$$H_\chi(s) = H(s) + 4\chi s(1-s)C_2, \tag{36}$$

where $\chi \in \mathbb{R}$ is the dimensionless catalyst strength. The operator C_2 commutes with both S_k and G_B, and its envelope vanishes at $s = 0, 1$. At $\chi = 2$, a crossing between different D_4 irreps occurs at $s \simeq 0.806454$, while the D_4-trivial gap there is approximately 0.258. This diagnostic is not used in the asymptotic theorem. Appendix C.2 specifies its numerical protocol, minimum-gap scan, and residuals.

4.2 An exact multiplicity closure on the even-cycle family

The asymptotic witness is analytic. Express minimum vertex cover on the even cycle C_n, $n = 2k$, as set cover: bases are vertices, tasks are edges, and each base covers its two incident edges. The only minimum covers are the two alternating subsets, which form one orbit of D_n. Here D_n denotes the dihedral group generated by the rotations and reflections of C_n.

For this family $n_B = n$. Consider the original ordered-register linear path $H(s)$ in Eq. (8). The off-diagonal coefficient of each single-register complete-graph move is

$$-\frac{1-s}{n} + \frac{s}{n-1}, \tag{37}$$

and vanishes exactly at

$$s_c = \frac{n-1}{2n-1}. \tag{38}$$

At this point the Hamiltonian is diagonal up to a scalar. All $2k!$ ordered representatives of the two alternating covers are global ground states, whereas their normalized orbit is the unique ground vector in the $S_k \times D_n$ fixed space. Assuming $\mu \ge \lambda$, every other orbit has diagonal penalty at least λ. This lower bound is attained: the distinct-base subset

$$S_\star = \{1\} \cup \{2j : 1 \le j \le k-1\} \tag{39}$$

has exactly one uncovered edge and no duplicate. Its joint orbit state therefore lies exactly λs_c above the ground energy. Write g_0^{global} and g_0^{sym} for the ranks of the global and joint-fixed ground projectors, and write Δ_{sym} for the joint-fixed excitation gap $\Delta_{\mathrm{exc}}^{\mathcal{H}_{\mathrm{sym}}}$ defined in Section 3.1. Consequently,

$$g_0^{\mathrm{global}}(s_c) = 2k!, \qquad g_0^{\mathrm{sym}}(s_c) = 1, \tag{40}$$

$$\Delta_{\mathrm{sym}}(s_c) = \lambda\frac{n-1}{2n-1}. \tag{41}$$

Thus the global multiplicity gap closes exactly while the joint-fixed gap is bounded below by a constant. The multiplicity closure is dynamically dark for the uniform initial state and the symmetry-preserving path. No transversality of the coincident branches is asserted here.

Equation (41) controls the closure point, not the minimum gap on every part of the linear path. The next section introduces a different, explicit parent path on the same family and proves a uniform polynomial bound. Keeping these statements separate avoids turning a pointwise multiplicity-closure statement into an unsupported adiabatic claim.

5 Parent path and accessible gap

The distinction between a global gap and a dynamically accessible gap is not only a finite-instance effect. We now give an even-cycle family for which the symmetry reduction, the solution manifold, and an entire parent-Hamiltonian path can be controlled. The purpose of the family is a separation theorem, not a claim that vertex cover on a cycle is classically difficult.

5.1 Even-cycle set cover and its joint symmetry

Let the bases be the vertices of the even cycle C_n, $n = 2k$, and let the tasks be its edges. Base b covers the two edges incident on b. A k-subset $S \subset \mathbb{Z}_n$, where $\mathbb{Z}_n$ denotes the vertex labels modulo n, is therefore a cover precisely when it is one of the two alternating subsets. The two solutions are exchanged by a one-site rotation and form one orbit of the dihedral incidence automorphism group D_n, generated by cycle rotations and reflections.

We work in the injective, register-symmetric subspace, with basis $\{|S\rangle : |S| = k\}$. Equivalently, this is the symmetric Dicke embedding of the hard-core sector of the original registers. Define

$$U(S) = |\{i \in \mathbb{Z}_n : i \notin S,\ i+1 \notin S\}|. \tag{42}$$

At half filling, $U(S)$ is also the number of adjacent occupied pairs. Thus $U = 0$ exactly on the two minimum covers.

For a dimensionless annealing parameter $a \ge 0$, consider the continuous-time Metropolis generator with a Johnson-graph proposal. For distinct k-subsets S, T, its off-diagonal transition rate is

$$q_a(S,T) = \frac{1}{n}\min\{1, n^{-a[U(T)-U(S)]}\}, \qquad |S\triangle T| = 2, \tag{43}$$

and $q_a(S,T) = 0$ otherwise; $|S \triangle T| = 2$ means that one selected vertex is replaced. Its reversible measure is

$$\pi_a(S) = Z_a^{-1} n^{-aU(S)}. \tag{44}$$

Here $Z_a = \sum_{|S|=k} n^{-aU(S)}$ normalizes the probability distribution. The symmetric-discriminant construction is standard for reversible Markov chains and Gibbs-amplitude ground states [9, 10]. Here H_a has diagonal entry $\langle S|H_a|S\rangle = \sum_{T\neq S} q_a(S,T)$ and, for $|S \triangle T| = 2$, off-diagonal entries

$$\langle T|H_a|S\rangle = -\sqrt{q_a(S,T)q_a(T,S)} \tag{45}$$
$$= -\frac{1}{n} n^{-a|U(T)-U(S)|/2}. \tag{46}$$

All remaining off-diagonal entries are zero, so these equations define H_a completely. It is frustration free and has the known ground state

$$|\psi_a\rangle = Z_a^{-1/2} \sum_{|S|=k} n^{-aU(S)/2}|S\rangle. \tag{47}$$

At $a = 0$, this is the Dicke state $|J_{n,k}\rangle = \binom{n}{k}^{-1/2} \sum_{|S|=k}|S\rangle$, and the parent Hamiltonian $H_{a=0}$ is exactly the Johnson-graph Laplacian at rate $1/n$. It should not be confused with the restriction of H_{init} from Eq. (7), which agrees with a Johnson Laplacian only up to an additive constant after projection to the injective subspace. The whole parent path commutes with D_n.

This protocol has its own generator algebra and cyclic space, distinct from Eq. (16):

$$\mathfrak{A}_{\text{par}} = \text{alg}^*(\{H_a : 0 \le a \le 7\} \cup \{I\}), \tag{48}$$
$$\mathcal{K}_J^{\text{par}} = \mathfrak{A}_{\text{par}}|J_{n,k}\rangle. \tag{49}$$

Here I is the identity on the half-filled hard-core space. Let $\mathcal{H}^{D_n}$ denote the D_n-fixed subspace of the half-filled hard-core space. Define the fixed-space and cyclic gaps by

$$\Delta_{D_n}(a) = \text{gap}(H_a|_{\mathcal{H}^{D_n}}), \tag{50}$$
$$\Delta_{\text{par}}^{\text{cyc}}(a) = \text{gap}\Big(H_a|_{\mathcal{K}_J^{\text{par}}}\Big). \tag{51}$$

Proposition 4 (Cyclic accessibility of the parent path)**.** *For every even $n = 2k \ge 6$ and $0 \le a \le 7$,*

1. *$\mathcal{K}_J^{\text{par}}$ is a common reducing subspace for the family $\{H_a\}_{0\le a\le 7}$ and $\mathcal{K}_J^{\text{par}} \subseteq \mathcal{H}^{D_n}$;*
2. *the unique ground state $|\psi_a\rangle$ belongs to $\mathcal{K}_J^{\text{par}}$; and*
3. $$\Delta_{\text{par}}^{\text{cyc}}(a) \ge \Delta_{D_n}(a). \tag{52}$$

Thus a lower bound proved in the D_n-fixed space is also a lower-bound certificate for the dynamically accessible gap of this parent protocol.

Physically, the protocol explores no more than the D_n-fixed sector, and restricting further to its true cyclic space cannot introduce an excitation below the fixed-sector gap when the unique ground state is shared. The full argument is in Appendix D.1.

The number of configurations at energy u is

$$N_u = \frac{n}{k-u}\binom{k-1}{u}^2, \qquad 0 \le u < k. \tag{53}$$

At $a_f = 7$, this shell count implies

$$1 - \langle\psi_{a_f}|\Pi_{U=0}|\psi_{a_f}\rangle = O(n^{-5}). \tag{54}$$

Here $\Pi_{U=0}$ is the orthogonal projector onto the two-dimensional span of the configurations satisfying $U(S) = 0$. Appendix D.2 gives the run decomposition and ratio bound.

5.2 A uniform accessible-gap theorem

We first separate the cycle geometry from the zero-range kinetics.

Proposition 5 (Zero-range reduction)**.** *For every even $n = 2k \ge 6$ and every $a \ge 0$, the gap of H_a in the D_n-fixed space obeys*

$$\Delta_{D_n}(a) \ge \frac{2}{n}[1-\cos(2\pi/k)]\,\text{gap}\,\text{ZRP}(K_k, r_a, k), \tag{55}$$

where $r_a(0) = 0$, $r_a(1) = n^{-a}$, $r_a(j) = 1$ for $j \ge 2$, and K_k is the uniform mean-field kernel $K_k(i,j) = 1/k$ on k boxes. The notation $\text{ZRP}(K_k, r_a, k)$ denotes the zero-range process with this jump kernel, rate function r_a, and k particles; its gap is the Poincaré constant of the corresponding generator. Consequently, any uniform polynomial lower bound for this mean-field gap gives a uniform polynomial accessible-gap certificate for the parent path.

The factors in Eq. (55) have distinct physical origins: $2/n$ is the update-rate normalization, $1 - \cos(2\pi/k)$ is the long-wavelength transport cost of the cycle, and the zero-range gap measures the remaining occupation-redistribution bottleneck. The exact isometry, rate comparison, and normalization are proved in Appendix E.

Proposition 6 (Uniform accessible gap)**.** *For every even $n = 2k \ge 6$ and every $a \ge 0$,*

$$\text{gap}\,\text{ZRP}(K_k, r_a, k) \ge 2n^{-a}k^{-3}, \tag{56}$$
$$\Delta_{D_n}(a) \ge 1024\, n^{-(a+6)}. \tag{57}$$

Consequently, along $0 \le a \le 7$, $\min_a \Delta_{D_n}(a) \ge 1024\, n^{-13}$.

Physically, the polynomial bound accumulates three conservative costs: normalized subset updates, long-wavelength transport around the cycle, and redistribution of vacancies between effective boxes. Its exponent is a certificate, not a prediction of the finite-size gap scaling. The complete comparison is in Appendix E.

The gap theorem can be converted into an adiabatic statement in the abstract Hamiltonian-evolution model. This does not by itself supply a gate-level implementation of the path.

Corollary 7 (Conditional adiabatic preparation)**.** *In the abstract Hamiltonian-access model, let $a(s) = 7s$, $0 \le s \le 1$, and evolve from $|J_{n,k}\rangle$ under the restricted path $H_{a(s)}|_{\mathcal{K}_J^{\mathrm{par}}}$ for total time T. There is a universal constant $C > 0$ such that, for every $0 < \epsilon < 1$, the sufficient condition*

$$T \ge C\epsilon^{-1} n^{41} (\log n)^2 \tag{58}$$

produces a final state at distance $O(\epsilon)$ from $|\psi_7\rangle$, up to phase. Its cover probability is therefore $1 - O(n^{-5}) - O(\epsilon)$. This runtime excludes the costs of preparing the Dicke state and realizing or simulating $H_{a(s)}$.

Physically, the polynomial cyclic gap prevents dark global crossings from setting the preparation time of this protocol. The large runtime exponent instead reflects deliberately conservative derivative and inverse-gap bounds and does not include Hamiltonian-simulation or state-preparation costs. The complete adiabatic and measurement estimates are in Appendix D.3.

Exact D_n-orbit quotients for $n = 6, 8, \ldots, 18$ show finite-size gaps much larger than the conservative theorem, but are not used to infer scaling. The grid and all intermediate values are reported in Appendix F and the archived data.

The path also clarifies what is and is not inherited from the original multi-register Hamiltonian. Its endpoint ground state has the same solution-producing measurement objective, but the initial ground state is the injective Dicke state rather than the product-register uniform state, and the endpoint is the frustration-free Gibbs parent H_{a_f} rather than the original H_{prob}. No interpolation between these two endpoints is assumed. The exact dark multiplicity closure of the original linear interpolation and the uniformly polynomially gapped parent path are therefore complementary statements about the same symmetric instance family, not two descriptions of one protocol.

6 Numerical illustration and limitations

Table 1 shows three representative endpoint behaviours. The grid illustrates a nontrivial joint group and a global ground sector inaccessible from the uniform state. The random row shows the strict inclusion $\mathcal{K}_u \subsetneq \mathcal{H}_{\mathrm{sym}}$ without padding or base automorphisms. The final row has zero accessibility cost but an accessible excitation gap hundreds of times larger than the full gap. Hence neither equality nor inequality of endpoint ground energies identifies the dynamically relevant gap. All finite-instance Hamiltonians reported here and in Appendix F use the penalty energies $\lambda = 5$ and $\mu = 10$ in Eq. (6).

At the problem endpoint H_{prob}, write Δ_{full} for the excitation gap on $\mathcal{H}_{\mathrm{valid}}$ and Δ_{acc} for the excitation gap on $\mathcal{K}_u$. If $\mathcal{G}_{\mathrm{acc}}$ is the accessible ground eigenspace, define its worst-case cover probability by

$$p_{\mathrm{opt}}^{\min} = \min_{\substack{\psi \in \mathcal{G}_{\mathrm{acc}} \\ \|\psi\|=1}} \langle \psi | P | \psi \rangle, \tag{59}$$

where P is the feasible-cover projector from Section 3.

Across all eleven frozen instances, five have $\mathcal{K}_u \subsetneq \mathcal{H}_{\mathrm{sym}}$, nine have $\delta_{\mathrm{acc}} > 0$, and every accessible endpoint ground state is rank one with $p_{\mathrm{opt}}^{\min} \ge 0.9944$. These are finite diagnostics, not scaling evidence. Appendix F gives the full tables, residual-based degeneracy rule, irrep classification, and OR-Library provenance.

7 Discussion and conclusion

For a fixed initial state and a symmetry-preserving interpolation, the physically relevant spectrum is neither the full spectrum nor, in general, the entire group-fixed spectrum. It is the spectrum of the Hamiltonian restricted to the cyclic space generated by that protocol. Thus $\mathcal{K}_u \subseteq \mathcal{H}_{\mathrm{sym}}$ is the central distinction: changing the initial state, the generated algebra, or a preserved symmetry can change the accessible spectrum and therefore the applicable gap. The fixed-Hamiltonian no-go statement is correspondingly conditional, rather than independent of state preparation.

Within this framework, the sector-resolved localization theorem certifies cover-measurement probability while retaining the sector kinetic floor. Exact orbit quotients separate ground multiplicity, excitation gaps, isolation gaps, and accessibility costs, and expose coincidences between irreducible sectors that are dark to the symmetric protocol. On the even-cycle family, the original interpolation has an exact global multiplicity closure while the joint-fixed excitation gap remains constant. A distinct Johnson/Metropolis parent path has a uniformly polynomial cyclic gap bounded below by $1024\, n^{-13}$ along the path, and consequently a conditional polynomial adiabatic runtime in the abstract Hamiltonian-access model.

These results establish a structural separation, not a practical quantum speedup. The cycle instances are classically solvable, Dicke-state preparation has a cost that must be counted, and the Gibbs parent endpoint is not the original problem Hamiltonian. Likewise, the address count $k\lceil \log_2 n_B \rceil$ is not a complete hardware resource estimate: incidence access, flags, ancillas, and gate depth require a separate implementation analysis.

A natural next step is a decorated family whose classical cover structure is nontrivial while its joint orbit quotient, dark global branch, and polynomial accessible gap remain stable under controlled symmetry-preserving perturbations. More broadly, spectral claims for symmetric Hamiltonian algorithms should

Table 1: Representative problem-endpoint reconstructions. Dimensions and spectra are computed in the physical full Hamiltonian and exact cyclic restriction. The complete eleven-instance audit is in Appendix F.

instance	$\dim \mathcal{H}_{\mathrm{sym}}$	$\dim \mathcal{K}_u$	Δ_{full}	δ_{acc}	Δ_{acc}	$p_{\mathrm{opt}}^{\mathrm{min}}$
grid-2x4	22	22	0.000382	0.271	0.315	0.9944
rnd-b8-t7-p50-s002	120	101	0.0563	0.0581	0.128	0.9986
scpe1-b8-t30	792	792	0.00591	0	4.87	0.9949

identify not only a Hamiltonian and its global gap, but also the initial state, preserved symmetry, generated cyclic space, and isolated band actually followed by the protocol.

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Author contributions

Fabrício de Souza Luiz conceived the project, developed the mathematical analysis, implemented and validated the numerical calculations, and wrote the manuscript. Large-language-model tools from OpenAI were used to assist with editorial reorganization, language revision, bibliographic discovery, LaTeX editing, and the development and checking of reproducibility code. The author reviewed all model-assisted material, verified the mathematical and numerical results, and takes full responsibility for the content of the work.

Code and data availability

The frozen instance registry, source code, residual-based numerical checks, and machine-readable tables accompanying this preprint are available in the versioned repository `https://github.com/fsluiz/manuscript-encoding-symmetry-limits`. A version-specific Zenodo DOI will be added before journal submission. Original software is released under the MIT License; original data and documentation are released under CC BY 4.0. Third-party materials, including OR-Library incidence data and the journal class file, retain their source licenses and provenance.

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A Proof of sector localization

We prove Proposition 1 and record why its sector hypothesis cannot be silently replaced by cyclic restriction. The variational principle applied inside $\operatorname{Ran}\Pi_\gamma$ gives $E_0^\gamma \leq \eta_\gamma$. Combining this with Eq. (20) yields

$$\Gamma_\gamma \geq m - \kappa - \eta_\gamma, \tag{60}$$

which is Eq. (25).

For a normalized sector ground vector, write $p = P_\gamma\psi_\gamma$ and $q = Q_\gamma\psi_\gamma$. Since P, Q, and Π_γ commute, these are orthogonal components and the lower block of the eigenvalue equation is

$$(Q_\gamma H_{\text{prob}} Q_\gamma - E_0^\gamma)q = -B_\gamma p. \tag{61}$$

If $Q_\gamma = 0$, then $q = 0$ identically. Otherwise positivity of Γ_γ makes the compressed operator on the left invertible with inverse norm at most Γ_γ^{-1}. Therefore

$$\|q\| \leq \frac{\|B_\gamma\|}{\Gamma_\gamma}\|p\|. \tag{62}$$

Using $\|p\|^2 + \|q\|^2 = 1$ and solving for $\|q\|^2$ gives Eq. (24). Replacing Γ_γ by the positive lower bound $m - \kappa - \eta_\gamma$ weakens the right-hand side and proves the stated certificate.

For finite penalties, the proposition controls rather than eliminates leakage; exact support on covers requires an infinite-penalty limit or a special decoupling. Moreover, if $\Pi_\mathcal{K}$ denotes the orthogonal projector onto $\mathcal{K}_u$, it need not commute with P, so $P\Pi_\mathcal{K}$ need not be an orthogonal projector. The proposition therefore applies directly to genuine commuting symmetry sectors. Cyclic-space solution probability must be evaluated directly or bounded with a separately constructed commuting projector.

B Joint representation and orbit quotient

The incidence automorphism group G_A acts on the address Hilbert space only through its base projection. If $\rho(\pi_B, \pi_T) = \pi_B$, then elements with $\pi_B = \mathrm{id}$ form the kernel, and the faithfully represented group is $G_B \simeq G_A/\ker\rho$. Since the S_k register action commutes with the G_B base action, complete reducibility gives

$$\mathcal{H}_{\text{valid}} \simeq \bigoplus_{\alpha,\beta} V_\alpha^{S_k} \otimes V_\beta^{G_B} \otimes \mathcal{M}_{\alpha\beta}. \tag{63}$$

Every invariant Hamiltonian lies in the commutant and hence has the form in Eq. (13). The identity factors generate Schur multiplicities; equality of eigenvalues belonging to different (α, β) blocks is not enforced by this statement.

For completeness, let $\mathcal{O}$ and $\mathcal{O}'$ be computational-basis orbits of the joint action and choose $x \in \mathcal{O}$. Invariance gives

$$\langle\mathcal{O}'|H|\mathcal{O}\rangle = \frac{1}{\sqrt{|\mathcal{O}||\mathcal{O}'|}} \sum_{x'\in\mathcal{O}} \sum_{y\in\mathcal{O}'} H_{yx'} \tag{64}$$

$$= \sqrt{\frac{|\mathcal{O}|}{|\mathcal{O}'|}} \sum_{y\in\mathcal{O}'} H_{yx}, \tag{65}$$

which proves Eq. (18). The orbit states are therefore an exact orthonormal basis of the fixed space, not an effective approximation.

After quotienting by register permutations, basis vectors are weak multisets of size k over the bases. If $g \in G_B$ has $c_\ell(g)$ cycles of length ℓ, the trace of g on the symmetric tensor power $\mathrm{Sym}^k(\mathbb{C}^{n_B})$ is the coefficient

$$[z^k]\prod_{\ell\geq 1}(1 - z^\ell)^{-c_\ell(g)}. \tag{66}$$

Averaging this character over G_B projects onto the trivial irrep and gives Eq. (19). This counts $\mathcal{H}_{\text{sym}}$ only. Whether $|u\rangle$ is cyclic for the restricted endpoint algebra is a separate closure calculation.

C Stability and the finite D_4 diagnostic

C.1 Proof of the stability theorem

Simple isolated eigenvalues of a differentiable Hermitian family define differentiable local branches. Apply the implicit-function theorem to $F(s,\varepsilon) = E_a(s,\varepsilon) - E_b(s,\varepsilon)$ and use Eq. (33). This gives a unique nearby root $s_\star(\varepsilon) = s_\star + O(|\varepsilon|)$. Commutation with the symmetry projectors makes every inter-sector matrix element vanish, so the equality cannot become an avoided crossing. Finally, Weyl's bound moves each eigenvalue of the a block by at most $|\varepsilon|\|R(s)\|$. An internal spectral distance can shrink by at most twice this quantity, which proves the isolation-gap statement in Theorem 3.

C.2 Catalyst and numerical audit

For the frozen 2×4 incidence instance with $k = 3$, the faithful base group is D_4. The $S_3 \times D_4$ fixed quotient and the cyclic closure from $|u\rangle$ both have dimension 22, compared with 512 in the valid ordered space. The latter equality is an output of the closure calculation, not an assumption based on symmetry. The orbit-isometry, endpoint-invariance, and cyclic-invariance residuals are below 10^{-13}.

For the catalyst in Eqs. (35)– (36), set $\chi = 2$. The remaining global crossing is at $s \simeq 0.806454$. The two branches belong to different D_4 irreps, Eq. (32) has residual 4.7×10^{-14}, and the D_4-trivial gap there is approximately 0.258. The trivial-sector gap was evaluated on a uniform 2001-point grid and locally minimized in every grid bracket; its minimum is 0.0414 at $s \simeq 0.874587$. The crossing between the trivial floor and the orthogonal-complement floor was bracketed on 401 points and refined by a scalar root solve. These are residual-tested finite diagnostics, not interval-certified bounds, and are not used in the asymptotic theorem.

D Parent cyclic space and endpoint concentration

D.1 Cyclic accessibility

We prove Proposition 4. Because $\mathfrak{A}_{\mathrm{par}}$ is a unital star-algebra, $\mathfrak{A}_{\mathrm{par}}|J_{n,k}\rangle$ is invariant under every element of the algebra and its adjoint, hence is reducing. Every H_a commutes with D_n, and $|J_{n,k}\rangle$ is D_n-fixed, proving the first item.

All rates in Eq. (43) are positive on the connected Johnson graph. The zero eigenvalue of H_a is consequently simple, with ground vector Eq. (47). In finite dimension its spectral projector $P_a = |\psi_a\rangle\langle\psi_a|$ is a polynomial in H_a and belongs to $\mathfrak{A}_{\mathrm{par}}$. Positivity gives $\langle\psi_a|J_{n,k}\rangle > 0$, so $P_a|J_{n,k}\rangle$ is a nonzero multiple of $|\psi_a\rangle$. Finally, both restrictions have this unique ground state, while $\mathcal{K}_J^{\mathrm{par}} \subseteq \mathcal{H}^{D_n}$. Rayleigh–Ritz minimization over the smaller orthogonal complement proves Eq. (52).

D.2 Endpoint concentration

At energy u, occupied and vacant vertices decompose into the same number $k-u$ of positive cyclic runs. Choosing the marked start and the two run compositions gives

$$N_u = \frac{n}{k-u}\binom{k-1}{u}^2, \qquad 0 \le u < k, \tag{67}$$

which is Eq. (53). At $a_f = 7$, the $u = 1$ contribution is $N_1 n^{-7} = n(k-1)n^{-7} = O(n^{-5})$. Consecutive terms $N_u n^{-7u}$ have ratio at most $2k^2 n^{-7} = O(n^{-5})$. Summing the resulting geometric tail proves Eq. (54).

D.3 Adiabatic derivative bounds

A Johnson move removes one selected cycle vertex and inserts one unselected vertex, so $|U(T) - U(S)| \le 2$, and each subset has $k(n-k) = k^2$ neighbours. Differentiating the matrix elements in Eqs. (43) and (46), then bounding the operator norm by the maximum absolute row sum, gives, for constants $c_1, c_2 > 0$ independent of n,

$$\|\partial_s H_{7s}\| \le c_1 n \log n, \qquad \|\partial_s^2 H_{7s}\| \le c_2 n (\log n)^2, \tag{68}$$

Proposition 4 and Eq. (57) give a simple ground state and minimum cyclic gap at least $1024n^{-13}$.

Apply the quantitative gapped adiabatic bound [11] to $H_{7s}|_{\mathcal{K}_J^{\mathrm{par}}}$; restriction cannot increase either derivative norm. The three contributions scale as $n^{27}\log n$, $n^{27}(\log n)^2$, and $n^{41}(\log n)^2$, divided by T. The last dominates and proves Eq. (58). The final measurement statement follows from Eq. (54) and continuity of measurement probabilities in state distance.

E Normalized comparison proof for the parent-path gap

This appendix fixes the generator convention used in Propositions 5 and 6 and records each comparison factor. For a reversible generator $\mathcal{L}$ with stationary law ν, write

$$\mathcal{E}_{\mathcal{L}}(f,f) = -\langle f, \mathcal{L} f\rangle_\nu, \tag{69}$$

$$\mathrm{gap}(-\mathcal{L}) = \inf_{\mathrm{Var}_\nu(f)>0} \frac{\mathcal{E}_{\mathcal{L}}(f,f)}{\mathrm{Var}_\nu(f)}. \tag{70}$$

Lemma 8 (Three-box heat-bath contraction). *Let $w_0 = \tau$, $w_j = 1$ for $j \ge 1$, with $0 < \tau \le 1$. On weak compositions of any total m into three boxes, let one heat-bath step keep one uniformly chosen coordinate and resample the other two from their conditional* *product law. The continuous-time generator $\mathcal{L}^*_{3,m}$ obtained by subtracting the identity has gap at least $1/3$.*

E.1 Marked configurations and the cycle zero-range chain

Put $\tau = n^{-a}$ and let $\Omega_{n,k} = \{S \subset \mathbb{Z}_n : |S| = k\}$. Let $\mathcal{Q}_a$ denote the Markov generator with the rates in Eq. (43) and stationary law $\pi_a(S) \propto \tau^{U(S)}$; its symmetric discriminant is H_a. If D_{π_a} is diagonal with entries $\pi_a(S)$, then

$$H_a = -D_{\pi_a}^{1/2}\mathcal{Q}_a D_{\pi_a}^{-1/2}. \tag{71}$$

The unitary map $f \mapsto \sum_S \sqrt{\pi_a(S)} f(S)|S\rangle$ intertwines the D_n actions. Thus the gap of H_a in the fixed space is the Poincaré constant of $\mathcal{Q}_a$ on D_n-invariant observables.

Introduce the marked space

$$\widehat{\Omega}_{n,k} = \{(S,x) : S \in \Omega_{n,k},\ x \in S\}, \tag{72}$$
$$\widehat{\pi}_a(S,x) = \frac{\pi_a(S)}{k}. \tag{73}$$

Starting at the marked particle x, list the occupied vertices in cyclic order and let g_i be the number of vacant vertices after particle i. This gives a bijection

$$(S,x) \longleftrightarrow (x,g), \qquad x \in \mathbb{Z}_n, \tag{74}$$

between marked configurations and anchored weak compositions satisfying

$$g_i \geq 0, \qquad \sum_{i=1}^k g_i = k, \qquad U(g) = \#\{i : g_i = 0\}. \tag{75}$$

Changing the marked particle cyclically shifts g, while rotating or reflecting the subset changes only the anchor and the dihedral orientation. Consequently a D_n-invariant observable $f(S)$ lifts to a cyclic-and-reflection-invariant observable $F(g)$.

Define

$$r_a(0) = 0, \qquad r_a(1) = \tau, \qquad r_a(j) = 1 \quad (j \geq 2). \tag{76}$$

The product weight of a composition is

$$\prod_{i=1}^k \prod_{j=1}^{g_i} r_a(j)^{-1} = \tau^{-\#\{i:g_i>0\}} = \tau^{-k}\tau^{U(g)}. \tag{77}$$

Let ν_a be the probability law on weak compositions obtained by normalizing this product weight. The irrelevant factor τ^{-k} shows that the pushforward of the marked law after dropping the anchor is exactly the stationary composition law: every unanchored composition g has precisely n anchored preimages, all with the same weight. In particular,

$$\mathrm{Var}_{\pi_a}(f) = \mathrm{Var}_{\widehat{\pi}_a}(f) = \mathrm{Var}_{\nu_a}(F). \tag{78}$$

Let $P_{C_k}(i,j) = 1/2$ when $j = i \pm 1 \pmod k$ and zero otherwise, and let e_i be the ith coordinate vector. We use the Hermon–Salez convention

$$(\mathcal{L}_{P,r}F)(g) = \sum_{i,j} r(g_i)P(i,j)[F(g - e_i + e_j) - F(g)]. \tag{79}$$

The notation $\mathrm{ZRP}(P,r,k)$ below denotes the zero-range generator $\mathcal{L}_{P,r}$ restricted to configurations with k particles. Multiplication by $2/n$ makes every oriented cycle transfer occur at rate $r_a(g_i)/n$. Such a transfer is an adjacent Johnson move. If $g_i = 1$ and the receiving box is positive, it creates one new zero and has the Metropolis rate τ/n; if the receiving box is zero, its Metropolis rate is $1/n \geq \tau/n$. For $g_i \geq 2$, the move does not increase U and both rates equal $1/n$. Hence the full Johnson Dirichlet form dominates the local form and, under the lift above,

$$\mathcal{E}_{\mathcal{Q}_a}(f,f) \geq \frac{2}{n}\mathcal{E}_{\mathrm{ZRP}(P_{C_k},r_a,k)}(F,F). \tag{80}$$

The equality of variances is exact; the inequality comes only from dropping nonlocal Johnson edges and, in one local case, lowering a Metropolis rate.

Let $K_k(i,j) = 1/k$. The Hermon–Salez comparison theorem [13] states at the level of Dirichlet forms that

$$\begin{aligned}\mathcal{E}&_{\mathrm{ZRP}(P_{C_k},r_a,k)}(F,F) \\ &\geq [1 - \cos(2\pi/k)]\mathcal{E}_{\mathrm{ZRP}(K_k,r_a,k)}(F,F),\end{aligned} \tag{81}$$

because $1 - \cos(2\pi/k)$ is the Poincaré constant of P_{C_k} and that of K_k is one. Restricting the variational problem to dihedral-invariant F can only increase its infimum. Equations (78)–(81) prove Eq. (55) with its factor $2/n$.

E.2 Three-box contraction

Set $w_0 = \tau$ and $w_j = 1$ for $j \geq 1$. For $0 < z < 1$, the probability sequence $p_z(j) \propto w_j z^j$ is log-concave: the only nontrivial inequality is $w_1^2 \geq w_0 w_2$, which is precisely $\tau \leq 1$. Conditioning two independent variables with this law to have sum s removes the factor z^s and gives

$$\nu_s(u) = \frac{w_u w_{s-u}}{\sum_{v=0}^s w_v w_{s-v}}, \qquad 0 \leq u \leq s. \tag{82}$$

The discrete Efron monotonicity theorem [14] implies that $U_s \sim \nu_s$ is stochastically nondecreasing in s. Write $\preceq_{\mathrm{st}}$ for stochastic domination. Symmetry under $u \mapsto s - u$ gives, for $t = s + d$,

$$U_s \preceq_{\mathrm{st}} U_t \preceq_{\mathrm{st}} U_s + d. \tag{83}$$

The common-quantile coupling therefore obeys $0 \leq U_t - U_s \leq d$ almost surely. The two increments in the coupled splits are $U_t - U_s$ and $d - (U_t - U_s)$, so both are nonnegative.

On the three-box state space use $d_1(x,y) = \frac{1}{2}\sum_i |x_i - y_i|$. Couple two heat-bath steps by holding the same uniformly selected coordinate i. The two resampled pair totals differ by $d = |x_i - y_i|$, and the coupling above gives pairwise ℓ^1 distance d. Together with the held-coordinate distance, the post-update d_1 distance is at most $|x_i - y_i|$. Let W_{1,d_1} denote the 1-Wasserstein distance induced by d_1. Thus the discrete heat-bath kernel P satisfies

$$W_{1,d_1}(\mathsf{P}(x,\cdot),\mathsf{P}(y,\cdot)) \le \frac{1}{3}\sum_i |x_i - y_i| = \frac{2}{3}d_1(x,y). \tag{84}$$

For a function f, let $\mathrm{Lip}(f) = \sup_{x\neq y}|f(x) - f(y)|/d_1(x,y)$ be its Lipschitz seminorm. It follows that $\mathrm{Lip}(\mathsf{P}f) \le (2/3)\,\mathrm{Lip}(f)$. Every nonconstant eigenfunction on this finite metric space has positive Lipschitz seminorm, so all nonconstant eigenvalues of the reversible kernel have absolute value at most $2/3$. Therefore $\mathcal{L}^*_{3,m} = \mathsf{P} - I$ has gap at least $1/3$, uniformly in the conserved total m. This proves Lemma 8.

E.3 Caputo–Sasada recursion with the present normalization

For N boxes and conserved total m, let

$$\Omega_{N,m} = \{g \in \mathbb{Z}^N_{\ge 0} : \textstyle\sum_i g_i = m\}, \tag{85}$$

$$\nu_{N,m}(g) = Z^{-1}_{N,m}\prod_{i=1}^{N} w_{g_i}. \tag{86}$$

Here $Z_{N,m}$ is the normalizing partition function. Let E_{ij} denote conditional expectation under $\nu_{N,m}$ given all coordinates outside the unordered pair $\{i,j\}$. The complete-graph pair heat-bath generator is normalized as

$$\mathcal{L}^{N,m}_{\mathrm{HB}} = \frac{1}{N}\sum_{1\le i<j\le N}(E_{ij} - I). \tag{87}$$

For $N = 3$, the discrete kernel $I + \mathcal{L}^{3,m}_{\mathrm{HB}}$ keeps one uniformly chosen coordinate and resamples the remaining pair. It is exactly the kernel of Lemma 8, not a rescaled version.

The conditioned product measures above have the non-interference property required by the three-component argument. For $m = 0$ the conditioned law is a point mass, contributes no conditional variance, and its gap is assigned the convention $+\infty$. With

$$\underline{\lambda}_3 = \inf_{m\ge 1}\mathrm{gap}(-\mathcal{L}^{3,m}_{\mathrm{HB}}), \tag{88}$$

the lower-bound form of the Caputo–Sasada recursion [15, 16], in the normalization (87), is

$$\mathrm{gap}(-\mathcal{L}^{N,m}_{\mathrm{HB}}) \ge (3\underline{\lambda}_3 - 1)\left(1 - \frac{2}{N}\right) + \frac{1}{N}. \tag{89}$$

Lemma 8 gives $\underline{\lambda}_3 \ge 1/3$. The first term on the right of Eq. (89) is therefore nonnegative, and

$$\mathrm{gap}(-\mathcal{L}^{N,m}_{\mathrm{HB}}) \ge \frac{1}{N} \tag{90}$$

for every conserved total. In the application below, $N = m = k$.

E.4 Conditional two-box comparison

For the mean-field kernel $K_k(i,j) = 1/k$, decompose its zero-range generator into unordered pairs:

$$\mathcal{L}_{\mathrm{ZRP}(K_k,r_a,m)} = \frac{2}{k}\sum_{1\le i<j\le k}\mathcal{L}^{(2)}_{ij}. \tag{91}$$

Here $\mathcal{L}^{(2)}_{ij}$ is the conditional two-box zero-range generator with kernel K_2, whose two rows are $(1/2, 1/2)$. Thus a unit leaves either member of the pair toward the other at rate $r_a/2$. The prefactor $2/k$ in Eq. (91) converts this to the mean-field rate r_a/k.

Condition on pair total s. For $s \ge 2$, the stationary weights on $\{0,\ldots,s\}$ are τ at the two endpoints and one in the interior, with

$$Z_s = s - 1 + 2\tau. \tag{92}$$

Every nearest-neighbour edge conductance of $\mathcal{L}^{(2)}_{ij}$ is at least $\tau/(2Z_s)$. For the canonical path between $u < v$, Cauchy–Schwarz gives a factor $v - u \le s$. For every edge, the stationary masses on its two sides have product at most $1/4$. Let $\mathcal{E}_s$ be the Dirichlet form of this conditional two-box generator. Consequently

$$\mathrm{Var}_{\nu_s}(f) \le \frac{s}{4}\sum_{u=0}^{s-1}[f(u+1) - f(u)]^2 \le \frac{sZ_s}{2\tau}\mathcal{E}_s(f,f). \tag{93}$$

The conditional pair gap is at least

$$\frac{2\tau}{sZ_s} \ge \frac{2\tau}{s(s+1)} \ge \frac{\tau}{k^2}, \qquad 1 \le s \le k; \tag{94}$$

for $s = 1$ it equals τ, and $s = 0$ has no conditional variance.

Let $\mathcal{E}_{\mathrm{HB}}$ be the form of Eq. (87). Applying Eq. (94) conditionally in every pair and using Eq. (91) gives, with $\mathcal{V}_{ij}(f)$ denoting the expected conditional variance outside $\{i,j\}$,

$$\mathcal{E}_{\mathrm{ZRP}(K_k,r_a,k)}(f,f) \ge \frac{2}{k}\frac{\tau}{k^2}\sum_{i<j}\mathcal{V}_{ij}(f) \tag{95}$$

$$= \frac{2\tau}{k^2}\mathcal{E}_{\mathrm{HB}}(f,f). \tag{96}$$

Combining this with Eq. (90) yields

$$\mathrm{gap}\,\mathrm{ZRP}(K_k,r_a,k) \ge \frac{2\tau}{k^2}\frac{1}{k} = 2\tau k^{-3}. \tag{97}$$

This displays separately the pair-selection factor $2/k$, the conditional pair estimate τ/k^2, and the heat-bath gap $1/k$.

Finally, for $k \ge 3$, $1 - \cos(2\pi/k) \ge 8/k^2$. Substitution of Eq. (97) into Eq. (55) gives

$$\Delta_{D_n}(a) \ge \frac{2}{n}\frac{8}{k^2}\frac{2\tau}{k^3} = \frac{32\tau}{nk^5} = 1024\,n^{-(a+6)}, \tag{98}$$

because $k = n/2$ and $\tau = n^{-a}$. This completes the proof of Proposition 6. Writing $\Delta_{\mathrm{ZRP}}(a) = \operatorname{gap}\mathrm{ZRP}(K_k, r_a, k)$ and also using cyclic accessibility, the complete protocol-level comparison is

$$\begin{aligned}\Delta_{\mathrm{par}}^{\mathrm{cyc}}(a) &\geq \Delta_{D_n}(a)\\ &\geq \frac{2}{n}\left[1-\cos\left(\frac{2\pi}{k}\right)\right]\Delta_{\mathrm{ZRP}}(a)\\ &\geq 1024\, n^{-(a+6)}. \quad (99)\end{aligned}$$

F Numerical protocol and full tables

Inputs are frozen with identifiers and SHA-256 hashes. Every reported gap is computed either in the full Hamiltonian or in an explicitly labelled exact orbit quotient; projected PHP, Löwdin, and full-Hamiltonian values are never combined in one unlabeled column. Numerical degeneracy is accepted only relative to eigenpair residuals and matrix scale. Inside a degenerate manifold, let P_0 denote its spectral projector and let $\Pi_{\alpha\beta}$ project onto the joint irrep labelled by (α, β). The irrep content is obtained from

$$\operatorname{rank}(P_0 \Pi_{\alpha\beta} P_0), \quad (100)$$

not by assigning labels to solver-dependent individual eigenvectors.

In the tables below, g_0^{full} and g_0^{acc} are the ranks of the ground projectors on $\mathcal{H}_{\mathrm{valid}}$ and $\mathcal{K}_u$, respectively. A partition in square brackets labels the corresponding S_k irrep.

For sparse full-space calculations, the central class sum of register transpositions separates every S_k irrep for $k \leq 5$. The orbit of a lowest vector constructs its complete Specht multiplet; deflation then tests for further cospectral copies before g_0^{full} is assigned. For the grid row, the two-dimensional global ground manifold is the E irrep of D_4, whereas the accessible branch is A_1. At the analytic hopping-cancellation point, the joint-fixed ground rank agrees with the independently enumerated number of G_B-orbits of optimal covers in every row.

The largest orbit-invariance residual is below 5×10^{-13}, the largest cyclic-invariance residual below 2×10^{-9}, and every full-space eigenpair and class-sum residual used below is stored in the machine-readable output. Eight rows are subinstances extracted from the OR-Library Set Cover benchmarks [17, 18]; parent indices, extraction metadata, and hashes are stored in the instance registry.

For the parent family, exact D_n-orbit quotients for $n = 6, 8, \ldots, 18$ place the minimum of a 31-point grid at $a = 7$. Endpoint gaps decrease from 0.166670 at $n = 6$ to 0.0105026 at $n = 18$ and look approximately quadratic on these sizes. This is finite-grid evidence, not a scaling inference; Proposition 6 supplies the uniform certificate.

Table 2: Exact structural reduction for the frozen finite instances. All dimensions refer to the physical, unpadded address space.

instance	k	$\lvert G_B\rvert$	$\dim \mathcal{H}_{\text{valid}}$	$\dim \mathcal{H}_{\text{sym}}$	$\dim \mathcal{K}_u$	cover orbits
grid-2x4	3	8	512	22	22	2
rnd-b8-t7-p50-s002	3	1	512	120	101	12
rnd-b8-t8-p42-s014	4	1	4096	330	326	7
scpe1-b8-t20	4	1	4096	330	328	8
scpe1-b8-t30	5	1	32768	792	792	1
scpe2-b8-t20	4	1	4096	330	328	3
scpe2-b8-t30	5	1	32768	792	792	8
scpe3-b8-t20	3	1	512	120	119	2
scpe3-b8-t30	5	1	32768	792	792	4
scpe4-b8-t29	5	1	32768	792	792	1
scpe5-b8-t30	5	1	32768	792	792	2

Table 3: Problem-endpoint spectra reconstructed from the full Hamiltonian and the exact cyclic restriction. Here Δ is the gap above the entire ground manifold, and $p_{\text{opt}}^{\min}$ is minimized over normalized accessible ground states. The last column is the joint-fixed ground rank at the analytic hopping-cancellation point.

instance	ground S_k content	g_0^{full}	Δ_{full}	δ_{acc}	g_0^{acc}	Δ_{acc}	$p_{\text{opt}}^{\min}$	$g_0^{\text{sym}}(s_c)$
grid-2x4	$2[3]$	2	0.000382	0.271	1	0.315	0.9944	2
rnd-b8-t7-p50-s002	$[3] \oplus 2[2,1] \oplus [1,1,1]$	6	0.0563	0.0581	1	0.128	0.9986	12
rnd-b8-t8-p42-s014	$[2,1,1]$	3	0.000327	0.0274	1	0.0297	0.9955	7
scpe1-b8-t20	$[1,1,1,1]$	1	0.000539	0.0103	1	0.00178	0.9983	8
scpe1-b8-t30	$[5]$	1	0.00591	0	1	4.87	0.9949	1
scpe2-b8-t20	$[2,1,1]$	3	0.00137	0.00674	1	0.18	0.9966	3
scpe2-b8-t30	$[1,1,1,1,1]$	1	0.00118	0.0316	1	0.0829	0.9982	8
scpe3-b8-t20	$[2,1]$	2	0.000658	0.00499	1	0.256	0.9969	2
scpe3-b8-t30	$[3,1,1]$	6	0.000434	0.00864	1	0.247	0.9985	4
scpe4-b8-t29	$[5]$	1	0.00608	0	1	4.8	0.9945	1
scpe5-b8-t30	$[4,1]$	4	0.00306	0.00749	1	0.242	0.9962	2